Omer Sabri Emeksiz

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Education

Koç University Istanbul, Turkey
Master of Science in Computer Science and Engineering *October 2024 – June 2026 (Expected)*
GPA: 3.90/4.00

Marmara University Istanbul, Turkey
Bachelor of Science in Electrical Electronic Engineering - Honor *September 2021 – June 2024*
GPA: 3.33/4.00

Publications

Ö. S. Emeksiz, Y. Yemez, E. Erzin, T. M. Sezgin, "Early Prediction of Cybersickness from EEG in Immersive VR" , IEEE International Conference on Multimedia and Expo (ICME), 2026 (accepted)

Ö. S. Emeksiz, E. Maşazade, S. Selim, "AI-Driven Optimization of the Filling Process: A Comparison of Reinforcement Learning Methods" , IEEE 33rd Signal Processing and Communications Applications Conference (SIU), 2025

Ö. S. Emeksiz, E. Erzin, Y. Yemez, T. M. Sezgin, "A Convolutional Transformer Model for EEG-Driven Cybersickness Detection in VR Experience" , IEEE 33rd Signal Processing and Communications Applications Conference (SIU), 2025

Ö. S. Emeksiz, M. E. Ağcabay, E. Maşazade, S. Selim, T. Boysan, "A Reinforcement Learning Model for Industrial Filling Process Control" , IEEE 30th International Conference on Electronics, Circuits and Systems (ICECS), 2023

Work Experience

KUIS AI Center, Koç University Istanbul, Turkey
AI Researcher *October 2024 – Present*

- Conducting research on time series modeling and computer vision for EEG-based cybersickness in immersive VR.
- Conducting research on deep reinforcement learning and affective computing for facial expressions on social robots.
- Supervised by Prof. Metin Sezgin, Prof. Yücel Yemez, and Prof. Engin Erzin; member of IUI Lab and MVGL Lab.

Baykon Industrial Weighing Systems Istanbul, Turkey
AI Engineer *February 2024 – October 2024*

- Developed adaptive RL control policies (Monte Carlo, TD, Q-learning, MAB) achieving optimal coarse-to-fine feed switching for industrial filling process, leading and completing the TUBITAK–1501 industrial AI project.
- Collected an industrial filling dataset (1,000 filling runs) and deployed the system on an embedded filling machine.

TUBITAK Istanbul, Turkey
Engineer Intern *August 2023 – September 2023*

- Developed continuous time TCP/IP communication application using C# and data extraction tool using Python.

Open Business Software Solutions (OBSS) Istanbul, Turkey
Software Engineer Intern *July 2023 – August 2023*

- Developed a full-stack enterprise-level Human Resources Job Postings and Applications Management System web application by using Java, Spring Boot for backend and ReactJS, MUI for frontend.
- Used LDAP and LinkedIn API with Spring Security and JWT for authentication and authorization.
- Managed project with Apache Maven and utilized Hibernate, Spring Data JPA and MySQL for database.

Marmara University Istanbul, Turkey
Undergraduate AI Researcher *December 2022 – June 2024*

- Conducted research on reinforcement learning for industrial filling process control, formulating the MDP framework and optimizing feed switching policy using Monte Carlo methods, under the supervision of Prof. Engin Maşazade.

Kalem Software Istanbul, Turkey
Software Engineer Intern *August 2022 – September 2022*

- Developed backend services for two B2B client web applications: File Organizer, a file storage and management platform, and Material Waybill Organizer, a system for materials organization, using Java and Spring Boot.
- Managed project with Apache Maven and utilized Hibernate, Spring Data JPA and MySQL for database.

Ongoing Research

Autonomous Discovery of Facial Expressions on Social Robots using RL: Actor-critic RL framework that autonomously discovers diverse human-aligned facial expressions by exploring high-dimensional facial parameter spaces, validated on a physical social robot.

Wrap-Up!: AI-Powered Collaborative Meeting Visualization: Full-stack Chrome extension for Google Meet that transcribes live audio (STT), orchestrates an LLM to extract meeting structure, and renders a shared real-time canvas with proactive AI facilitation, thread tracking, and collaborative node interaction for all participants.

Multimodal EEG Dataset for Cybersickness in Immersive VR: Multimodal data collection study to establish an open benchmark for cybersickness research, addressing the lack of publicly available datasets in the field.

Research Interests

- Reinforcement Learning
- Time Series Modeling
- Computer Vision
- Affective Computing

Professional Service

Reviewer: ICME 2026, TMM, SIU 2026

Teaching: Teaching Assistant at Introduction to Computer Science and Programming, Computer Engineering Design, and Communication Engineering courses

Relevant Coursework

- Computer Vision
- Deep Learning
- Deep Unsupervised Learning
- Intelligent User Interfaces
- Medical Image Analysis
- Embodied Intelligence
- Biomedical Signal Processing
- AI in Health Sciences
- Image Processing

Skills

Programming Languages: Python, C, C++, Java

Libraries and Frameworks: PyTorch, NumPy, Scikit-learn, OpenCV, HuggingFace, Spring Boot, ReactJS